```
Tanzania, step 2 (2026-11-02)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.279             -43195.5            -43191.4                 0
     2 20263017             5000           0.276             -43089.7            -43088.4                 0
     3 20263018             5000           0.278             -43326.1            -43324.8                 0
     4 20263019             5000           0.276             -43266.3            -43262.3                 0
                                      file
 mcmc_tanzania_step2_chain1_2026-10-31.Rdata
 mcmc_tanzania_step2_chain2_2026-10-31.Rdata
 mcmc_tanzania_step2_chain3_2026-10-31.Rdata
 mcmc_tanzania_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
                          log.OR.engagement.slope     age.45.to.64.mortality.slope.multiplier.female
                                            3.082                                              3.021
            age.20.to.29.transmission.multiplier.1                     suppressed.disengagement.rates
                                            2.849                                              2.510
                    hiv.specific.mortality.rates.2            age.15.to.24.hiv.mortality.multiplier.0
                                            2.446                                              2.389
age.45.to.64.mortality.intercept.multiplier.female                          log.OR.suppression.slope
                                            2.236                                              2.136
            age.15.to.19.transmission.multiplier.2                                           trate.0
                                            2.068                                              2.065
           age.25.to.49.hiv.mortality.multiplier.0                                  age.assortativity
                                            2.039                                              1.990
                         log.OR.testing.intercept             age.40.to.49.transmission.multiplier.1
                                            1.955                                              1.926
            age.0.to.14.hiv.mortality.multiplier.2           age.25.to.49.hiv.mortality.multiplier.1
                                            1.871                                              1.806
            age.40.to.49.transmission.multiplier.3                      male.suppression.multiplier
                                            1.776                                              1.737
            age.30.to.39.transmission.multiplier.3                   male.hiv.mortality.multiplier.0
                                            1.718                                              1.695
                  male.hiv.mortality.multiplier.1            age.20.to.29.transmission.multiplier.2
                                            1.681                                              1.674
                       male.engagement.multiplier                over.50.hiv.mortality.multiplier.0
                                            1.655                                              1.633
            age.15.to.19.transmission.multiplier.1
                                            1.594

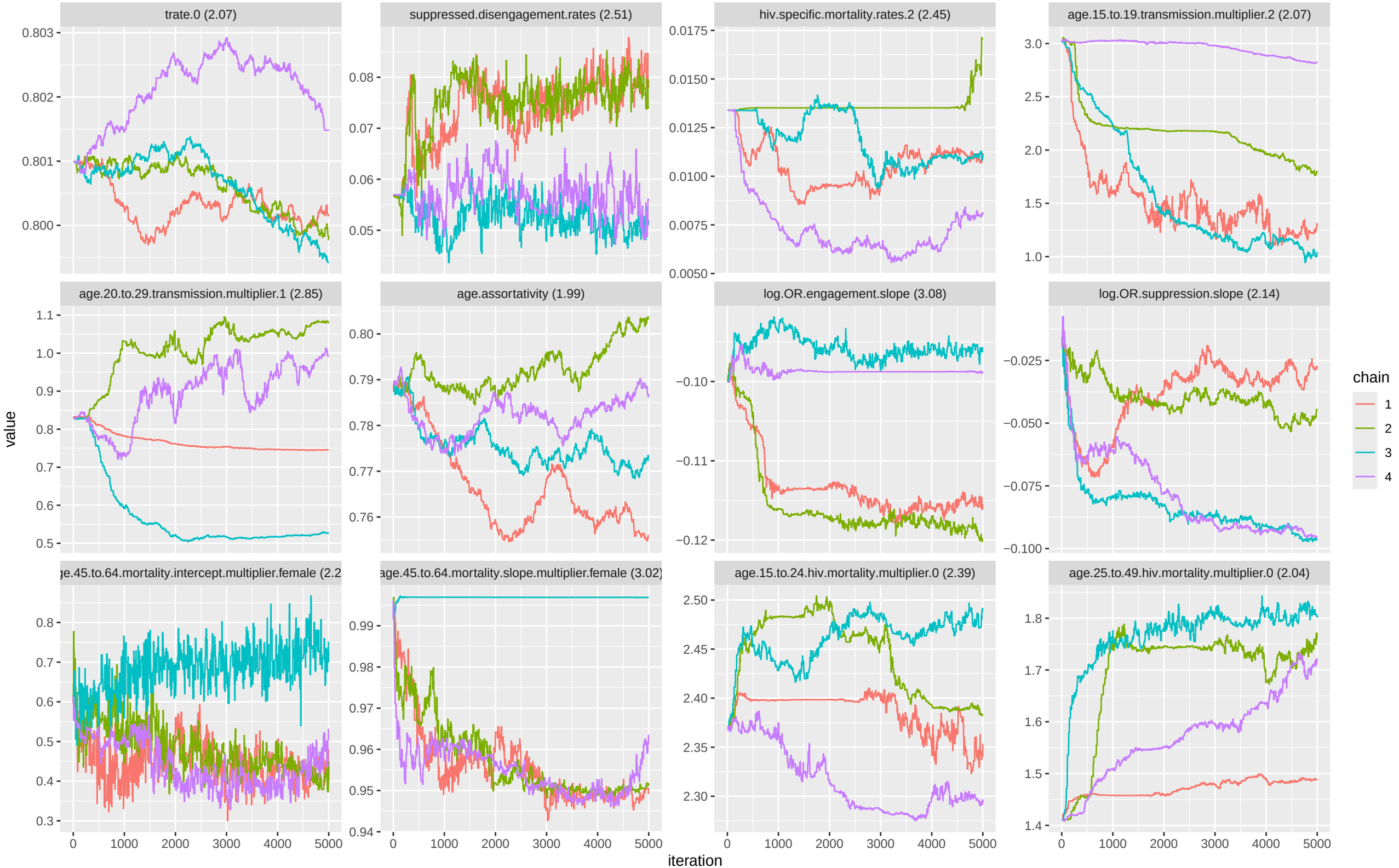
Merged simset:
 location simulations total.weight                                              file
 Tanzania        1000         1000 simset_tanzania_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 59 of 63 targets flagged (worst 30 shown; all in <location>_fit_check.csv)
     data.type         age    sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality       25-49 female              26                   0              x0.30      2025            x0.04
 hiv.mortality 50 and over   male              26                   0              x2.88      2025            x2.10
 hiv.mortality       15-24 female              26                   4              x0.40      2025            x0.21
 hiv.mortality       25-49   both              26                   0              x0.41      2025            x0.10
 hiv.mortality 50 and over   both              26                   0              x2.13      2025            x1.32
     incidence       10-19 female              26                   4              x0.51      2025            x0.17
 hiv.mortality       25-49   male              26                   8              x0.56      2025            x0.17
 hiv.mortality    All ages female              26                  15              x0.57      2025            x0.26
     incidence       15-24 female              26                   8              x0.59      2025            x0.27
```

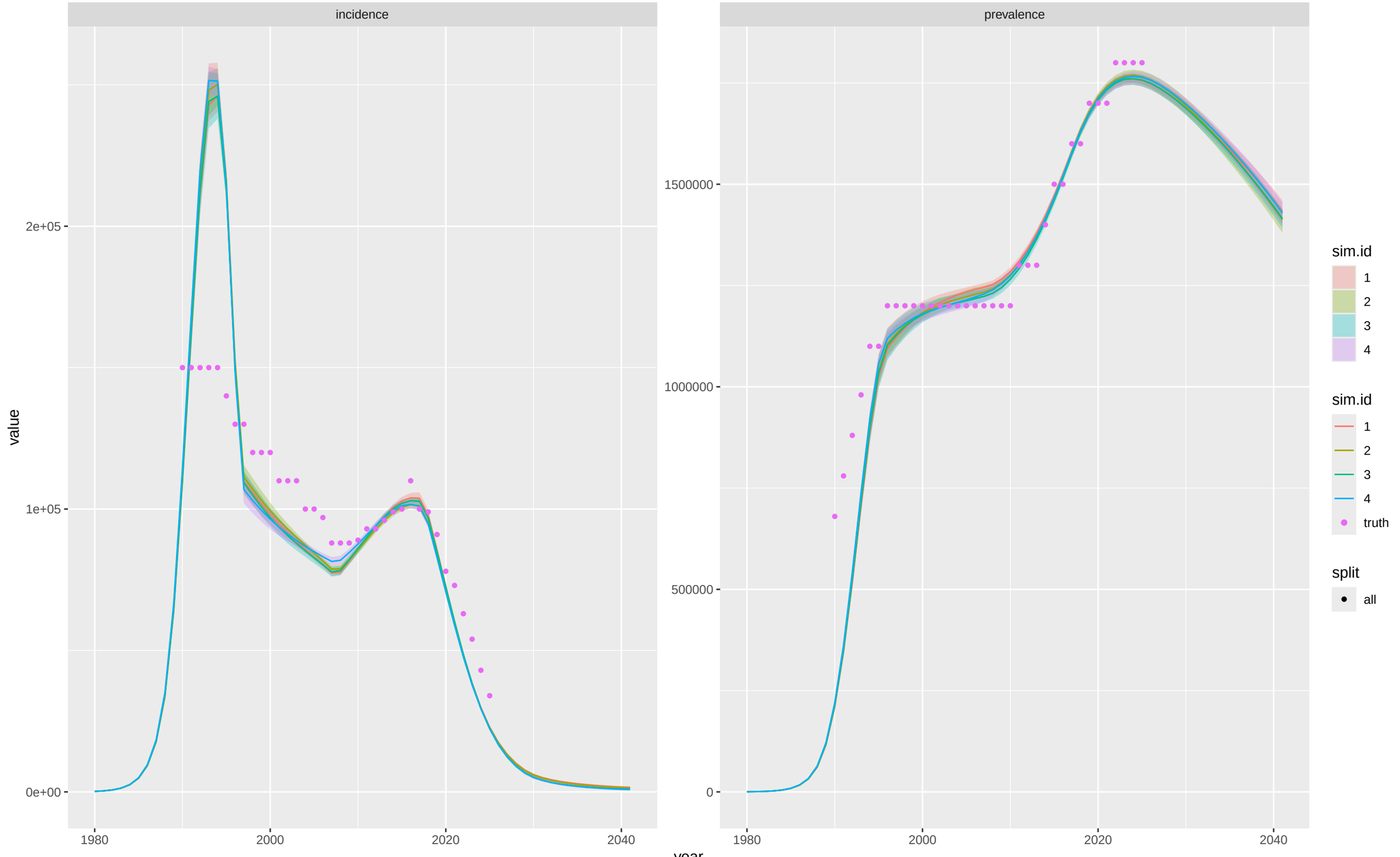
Tanzania step 2: likelihood

0
−10000
−20000
−30000
−40000
value
0
25000
50000
75000
100000
iteration
chain
1
2
3
4
type
Log Likelihood
Log Prior
Log Prior x Likelihood

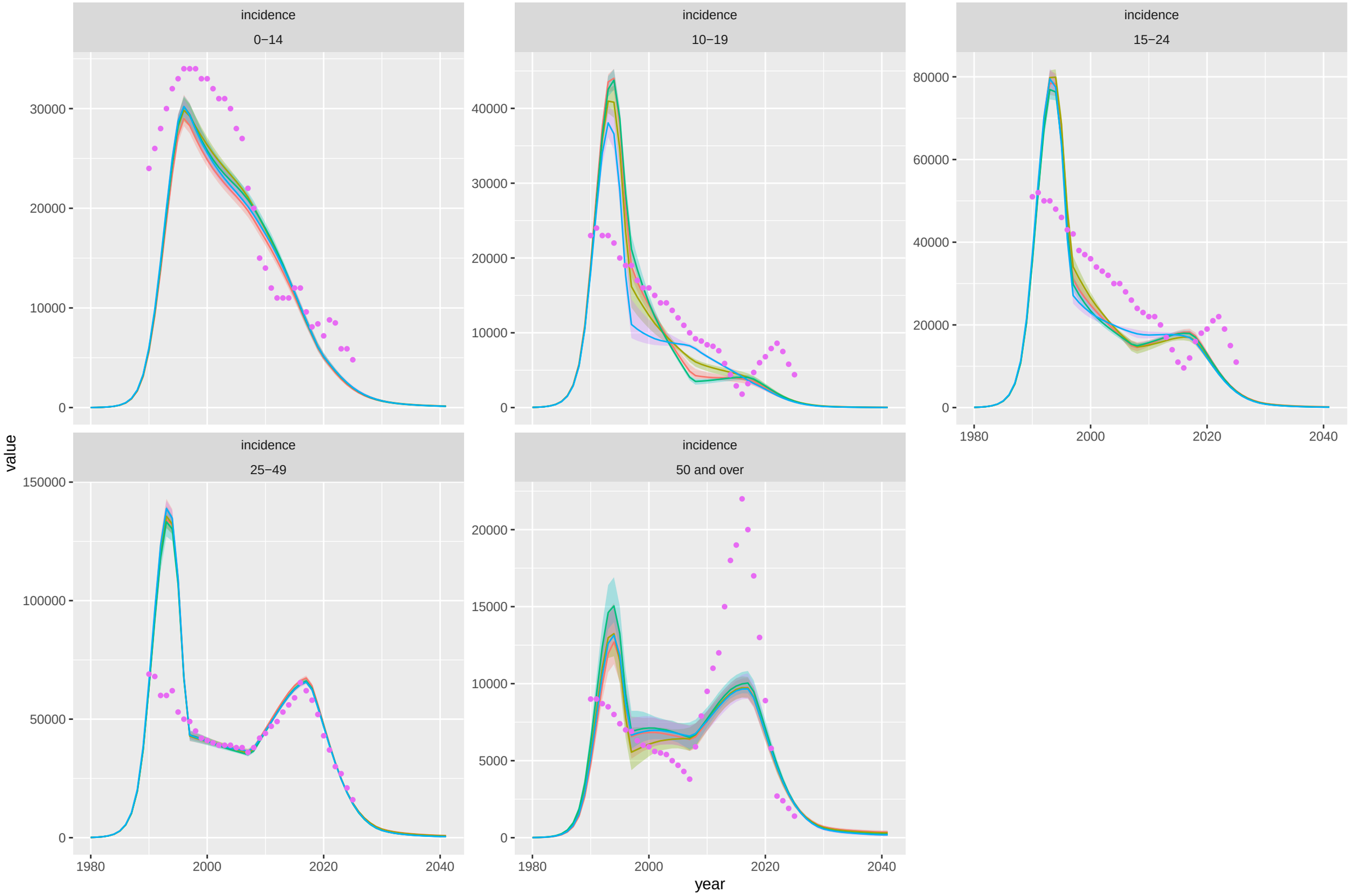
Tanzania step 2: trace



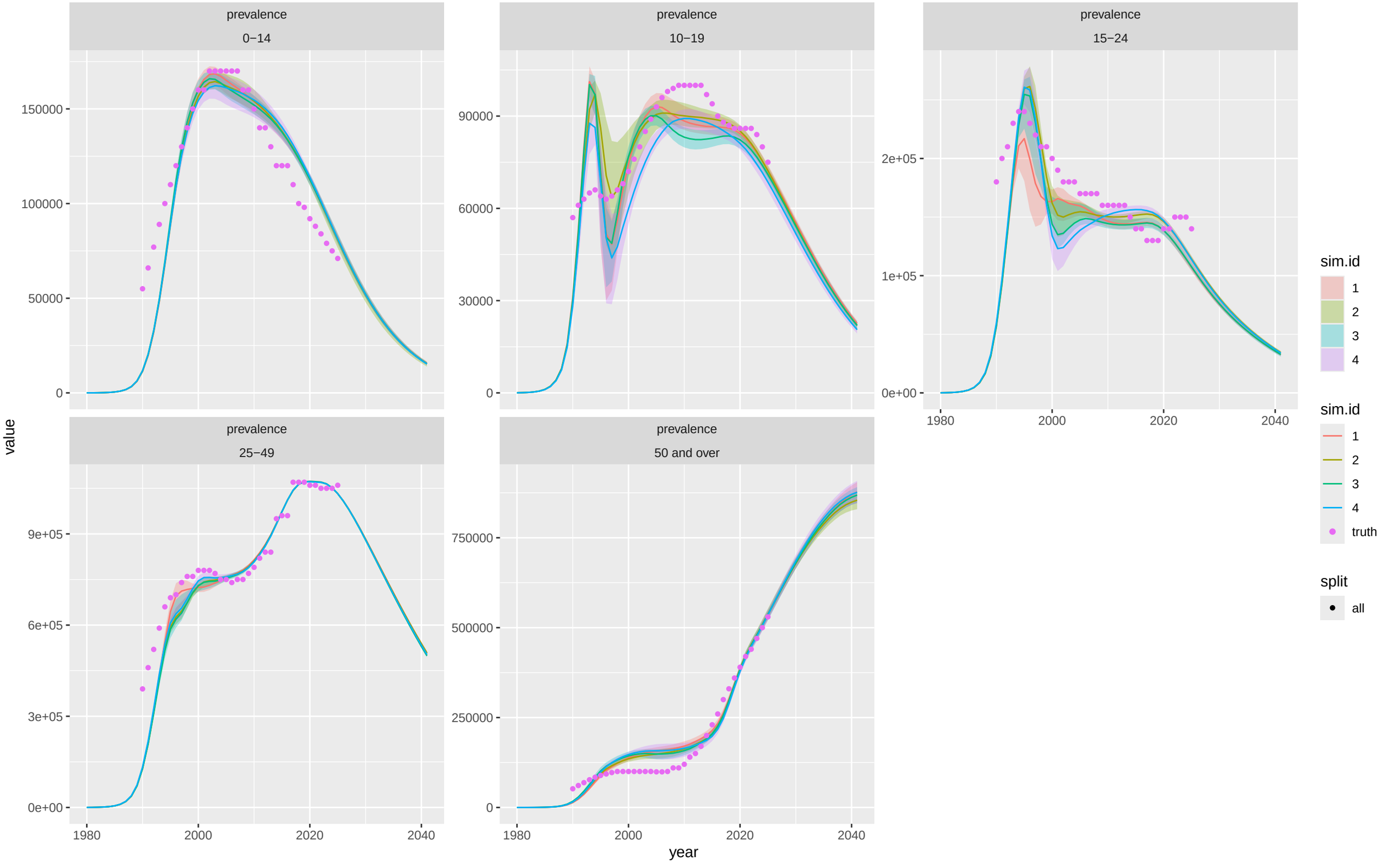
Tanzania step 2: fit (one color per chain)



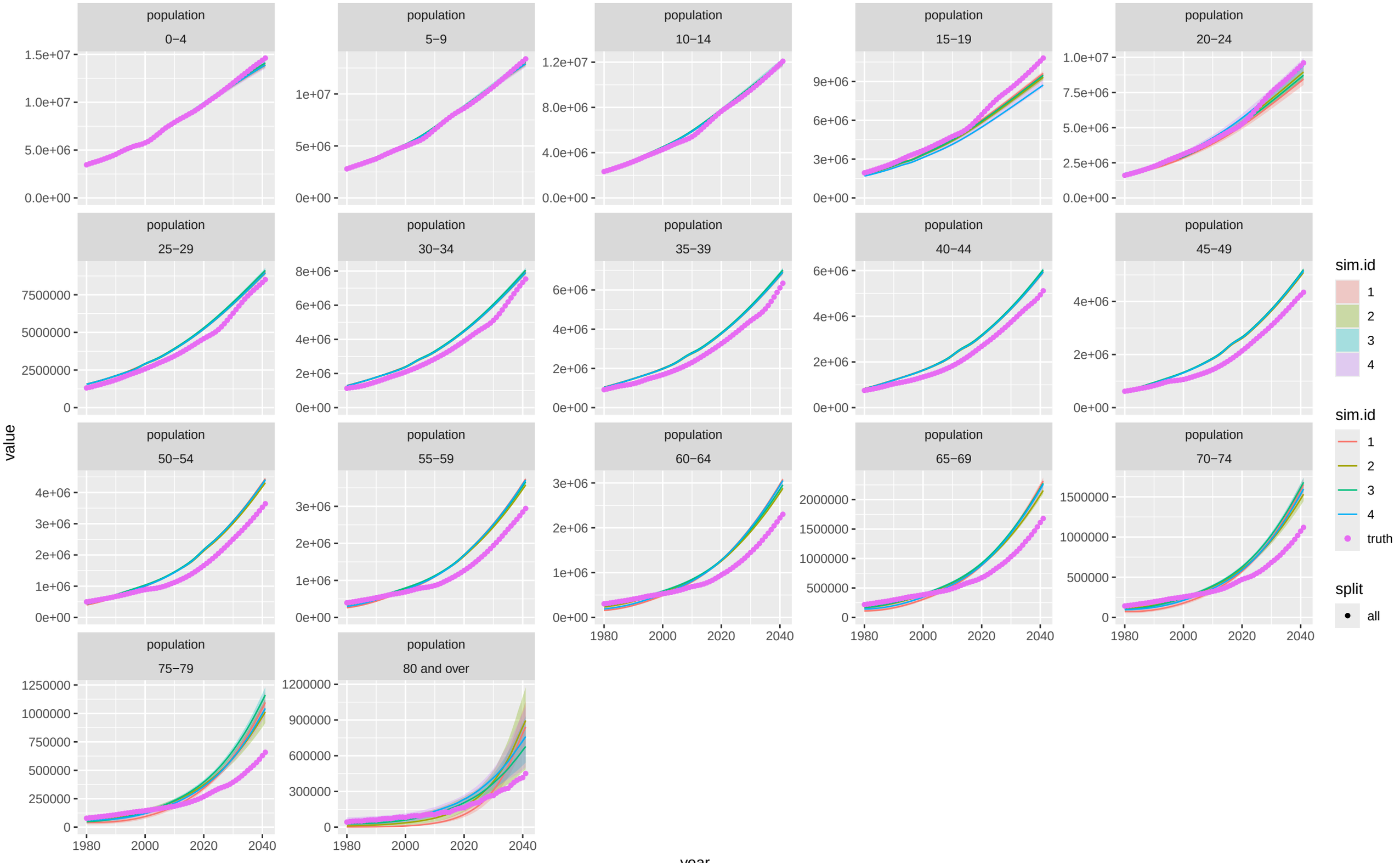
Tanzania step 2: incidence.by.age (one color per chain)



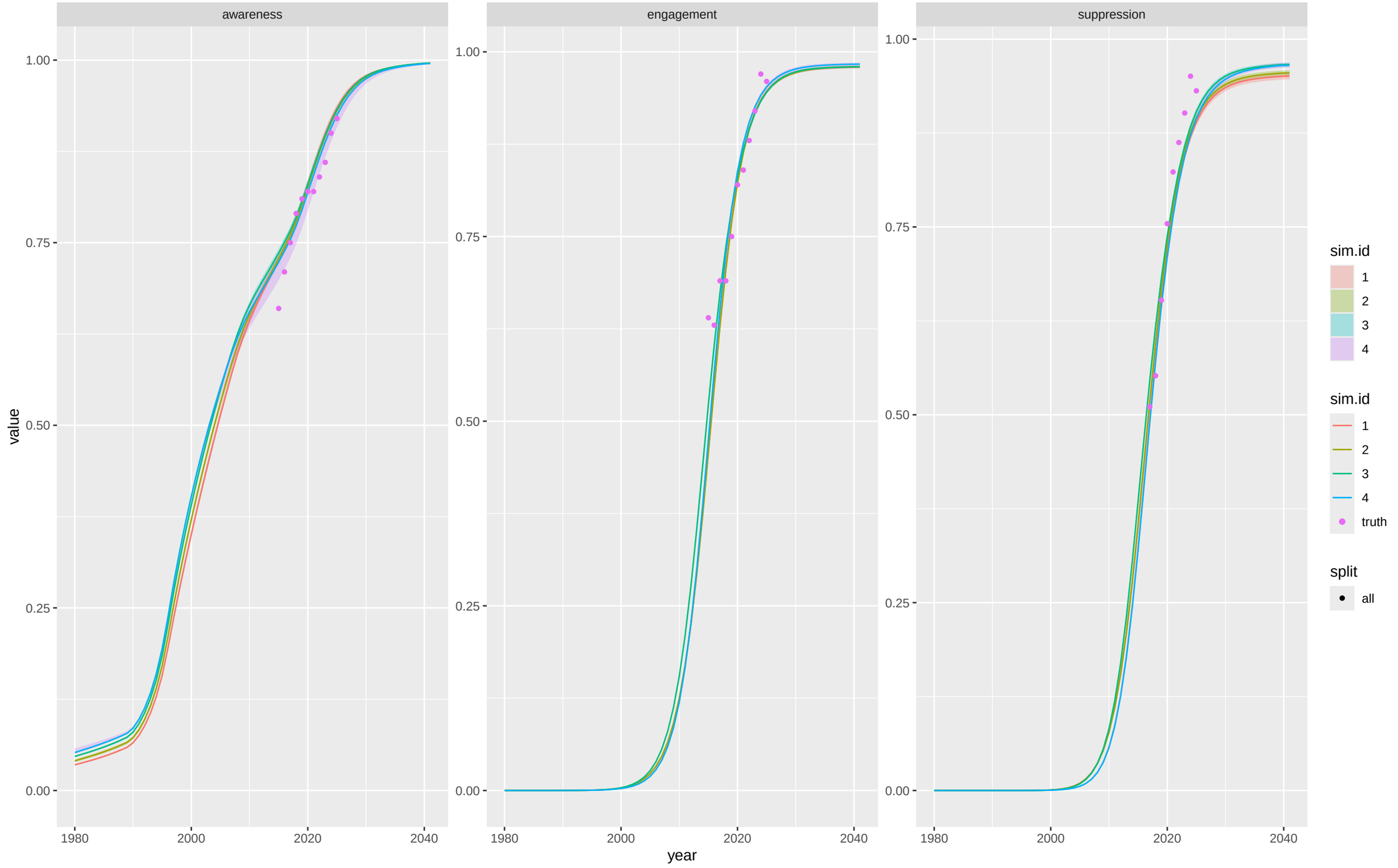
Tanzania step 2: prevalence.by.age (one color per chain)



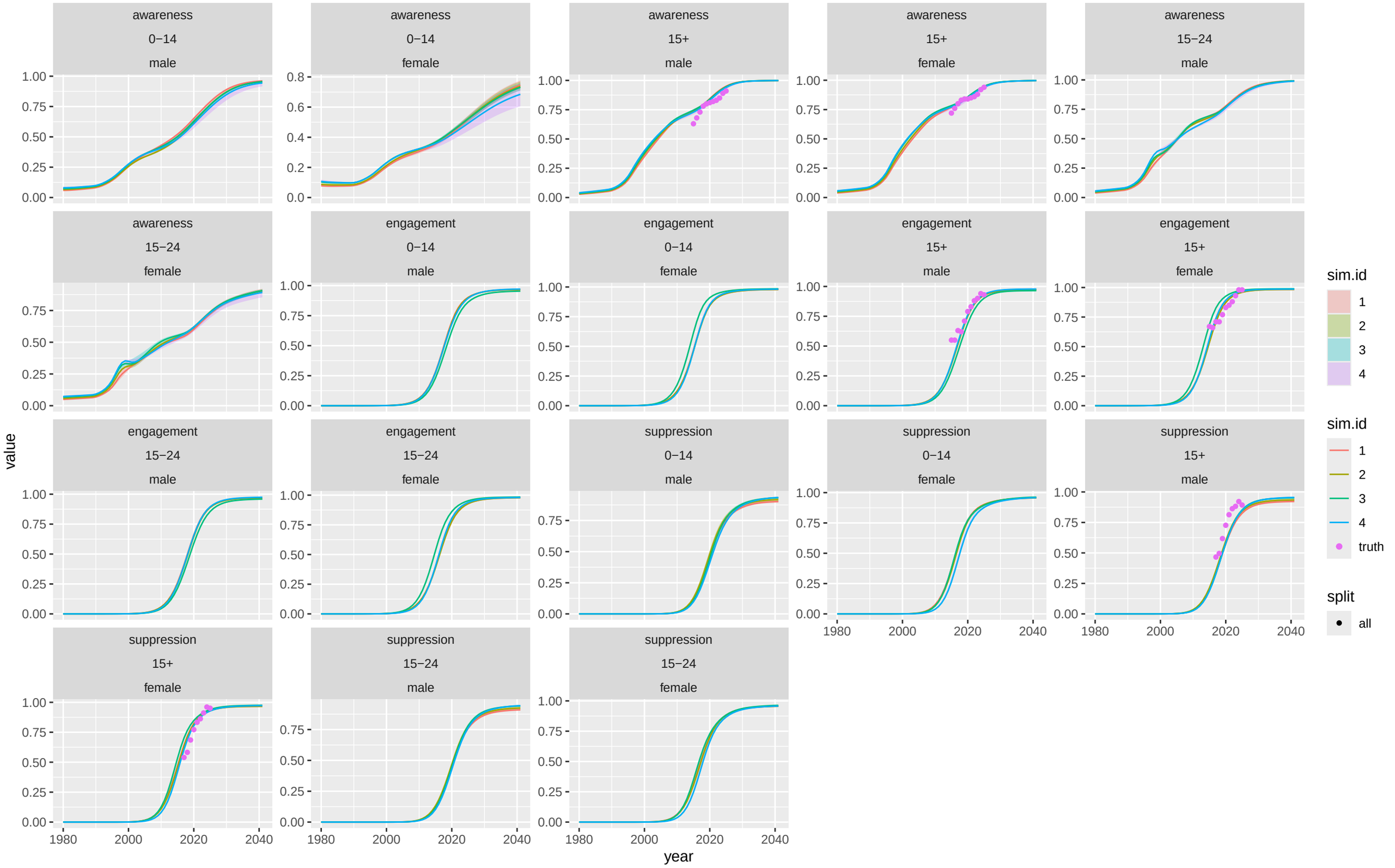
Tanzania step 2: population.by.age (one color per chain)



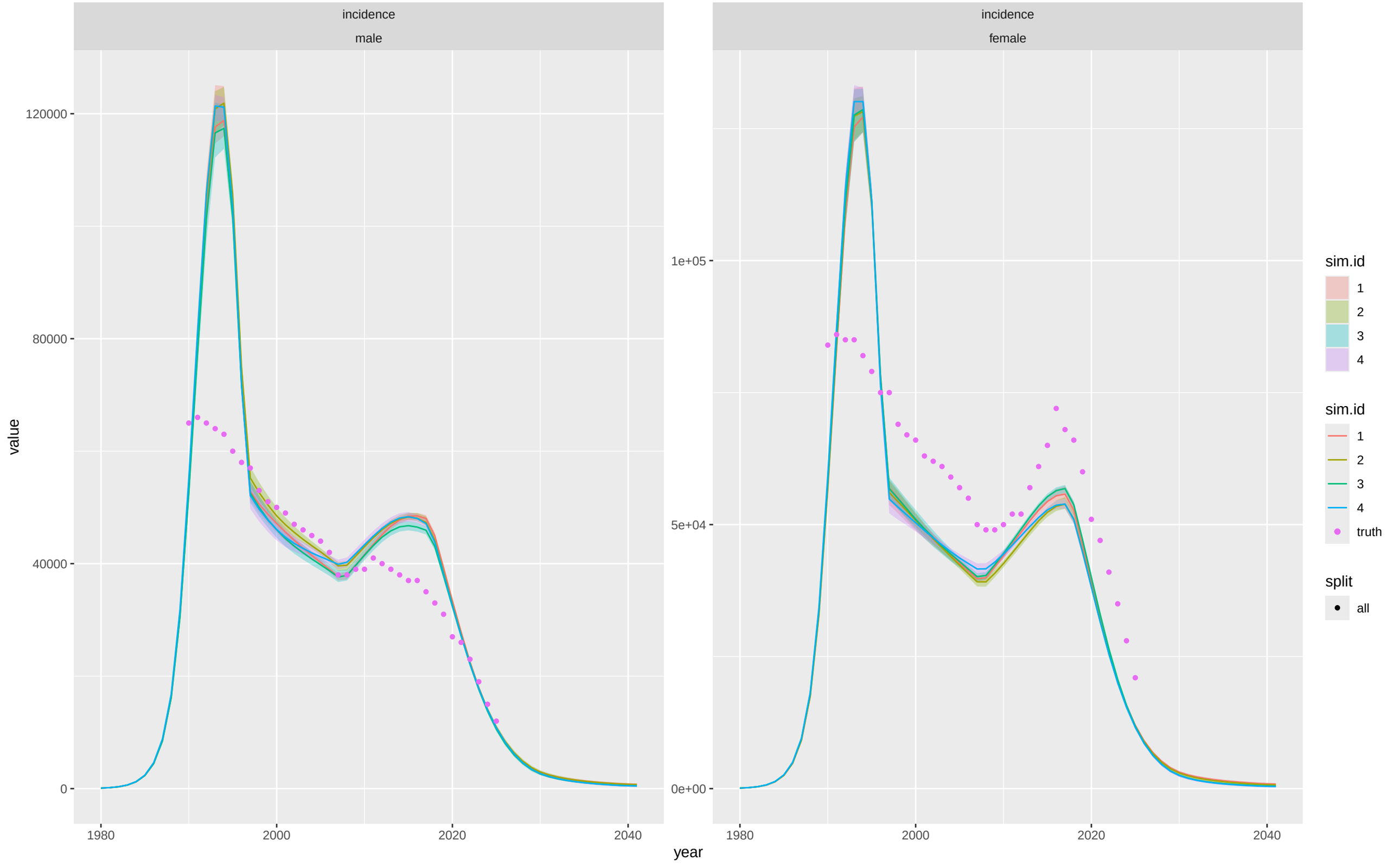
Tanzania step 2: cascade (one color per chain)



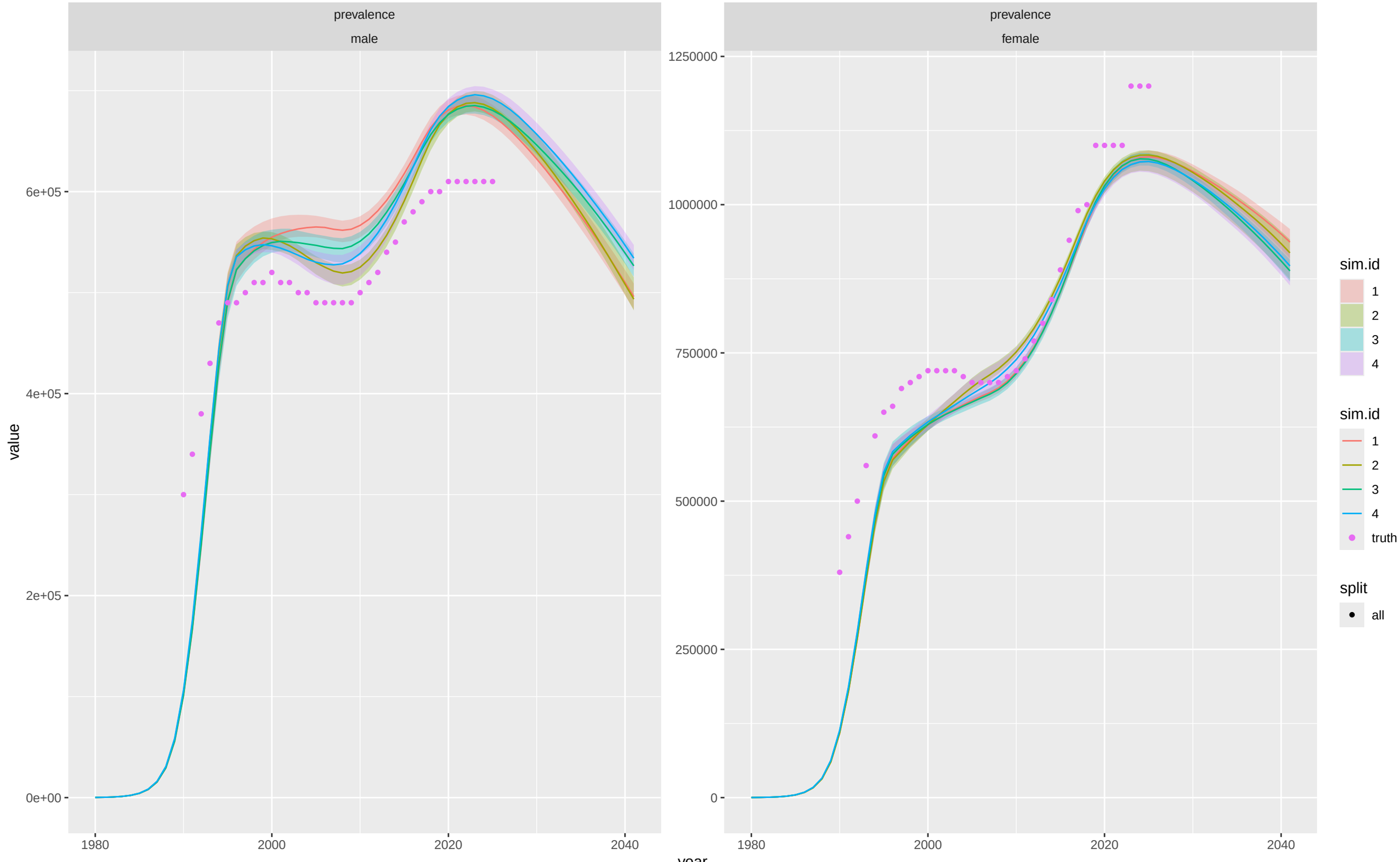
Tanzania step 2: cascade.by.age.sex (one color per chain)



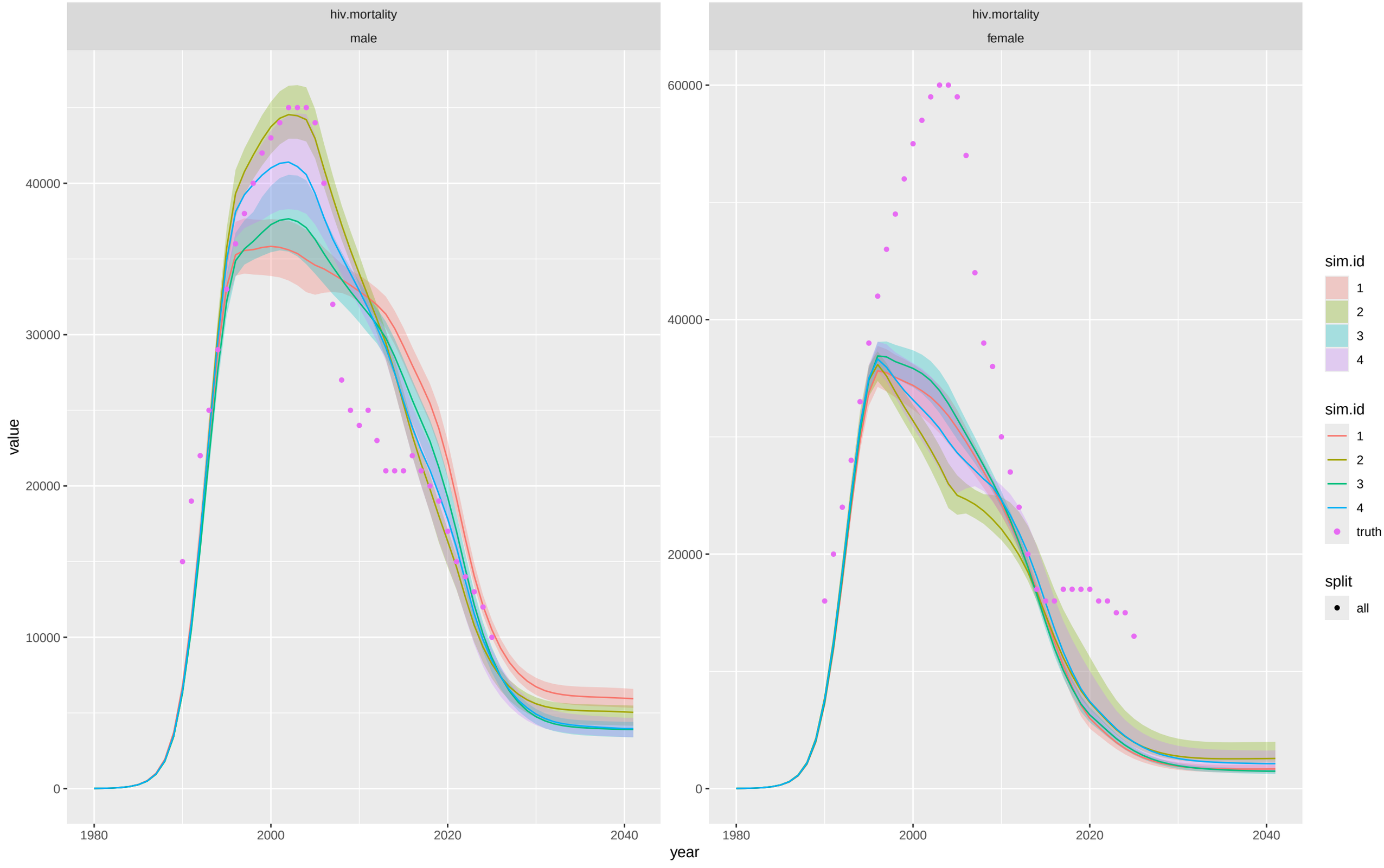
Tanzania step 2: incidence.by.sex (one color per chain)



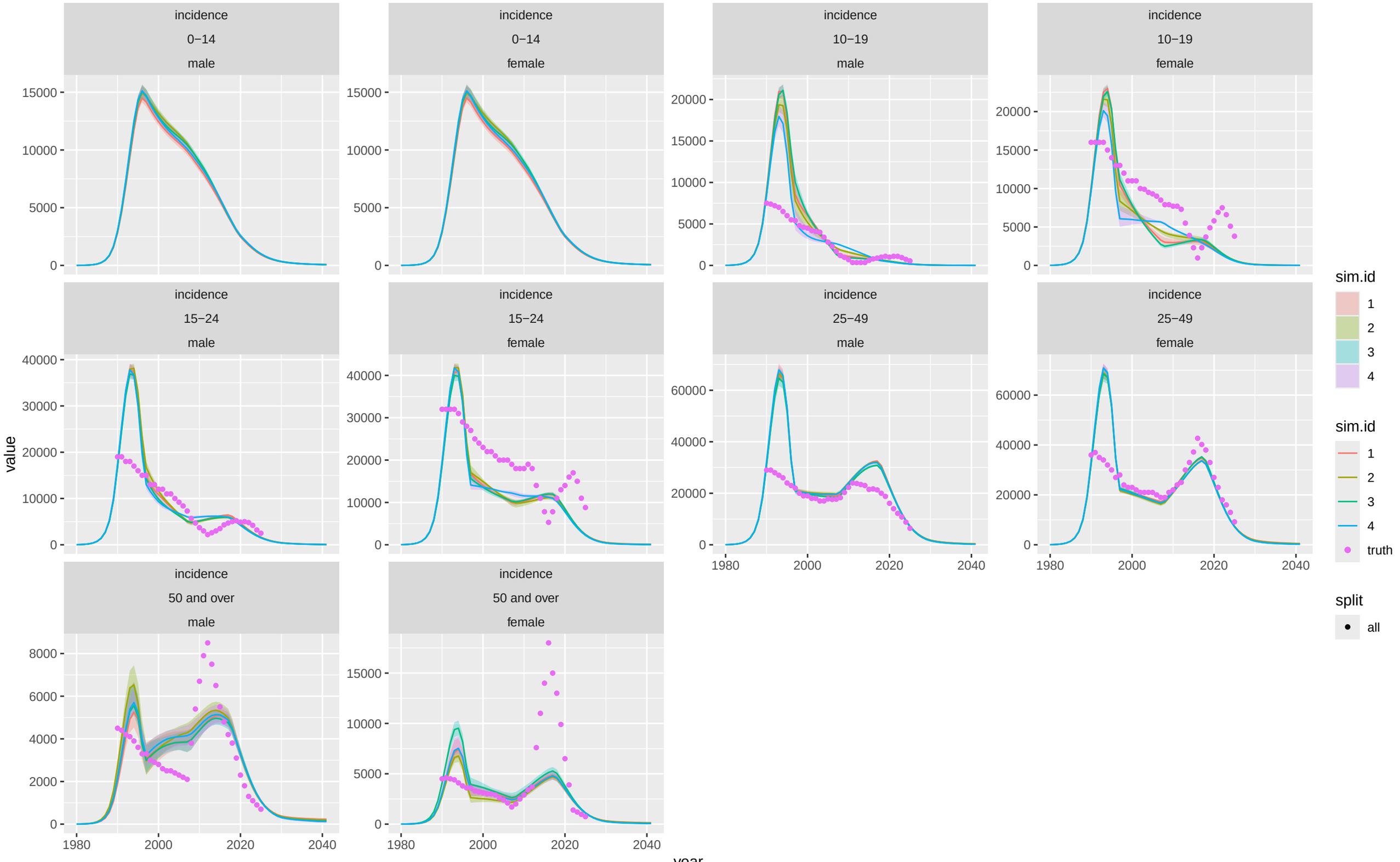
Tanzania step 2: prevalence.by.sex (one color per chain)



Tanzania step 2: hiv.mortality.by.sex (one color per chain)

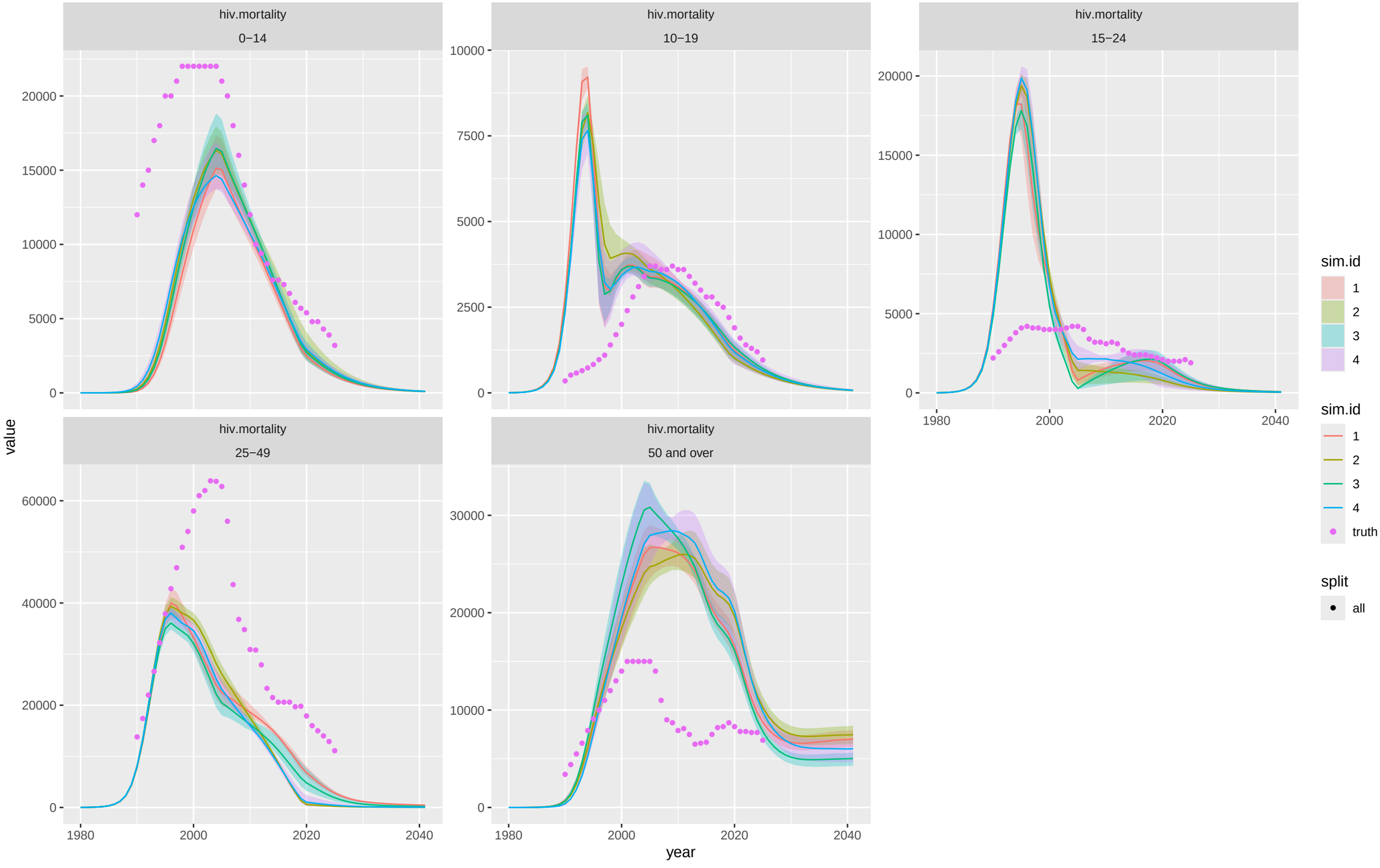


Tanzania step 2: incidence.by.age.sex (one color per chain)

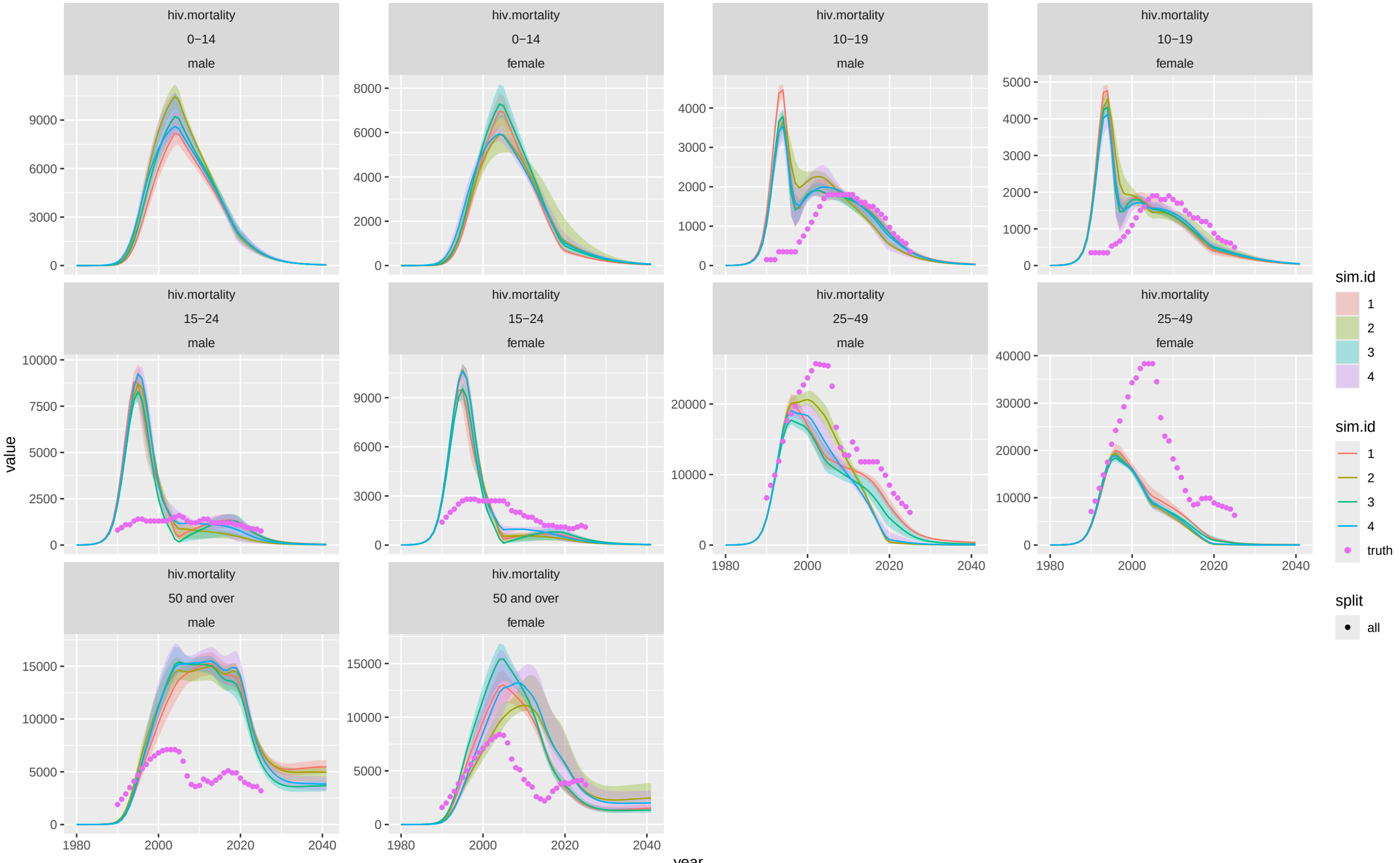


Tanzania step 2: prevalence.by.age.sex (one color per chain)

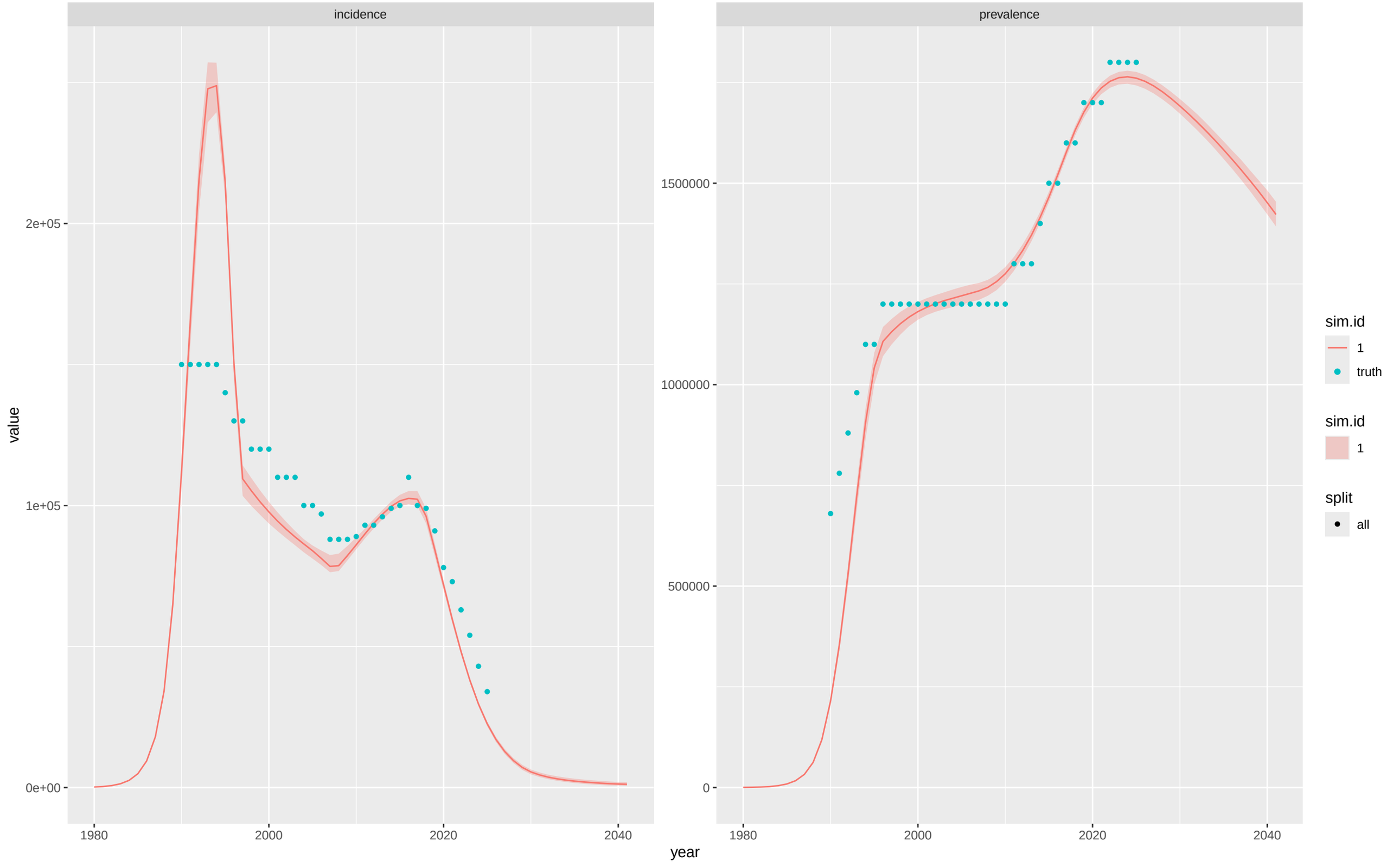
Tanzania step 2: hiv.mortality.by.age (one color per chain)



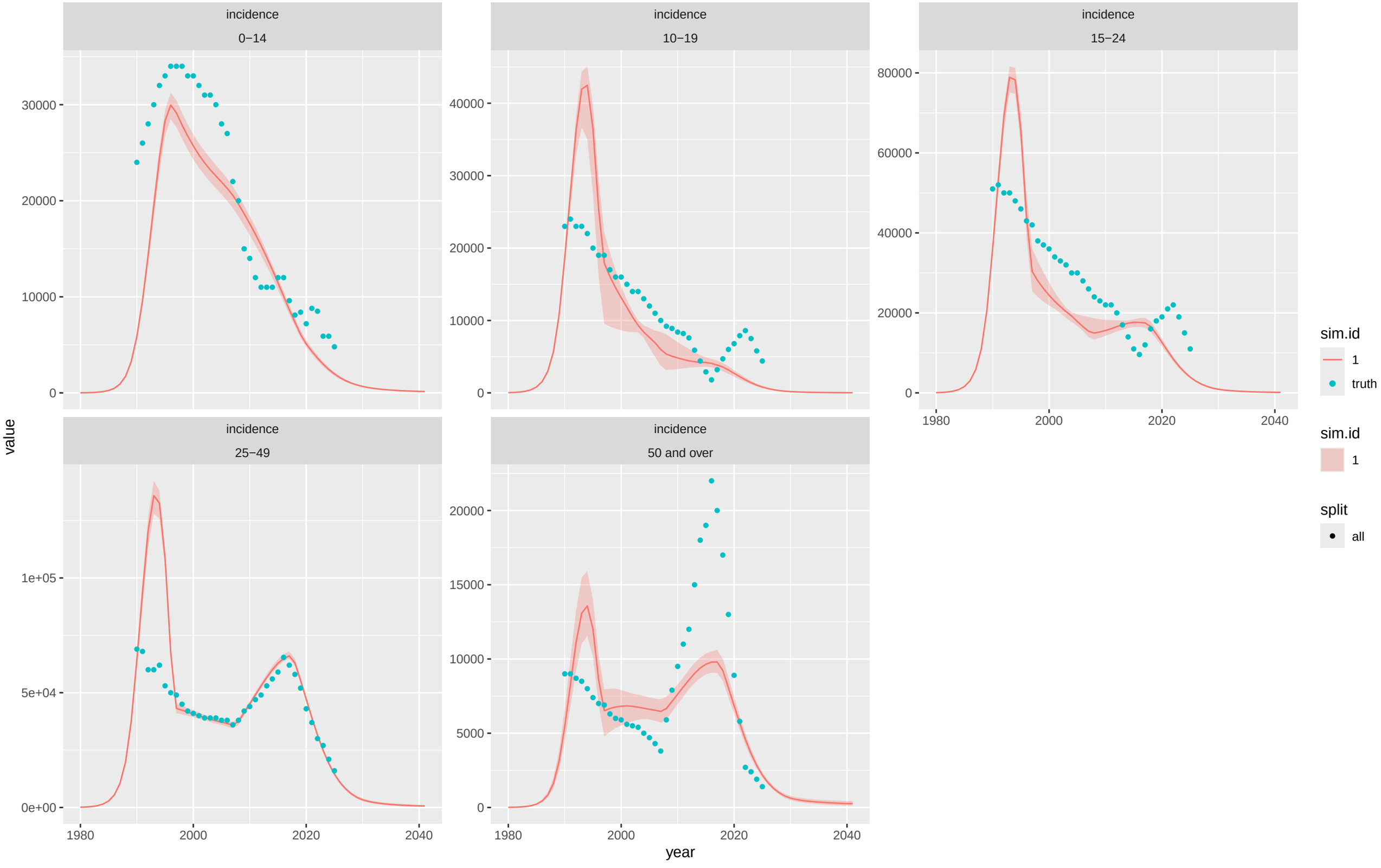
Tanzania step 2: hiv.mortality.by.age.sex (one color per chain)



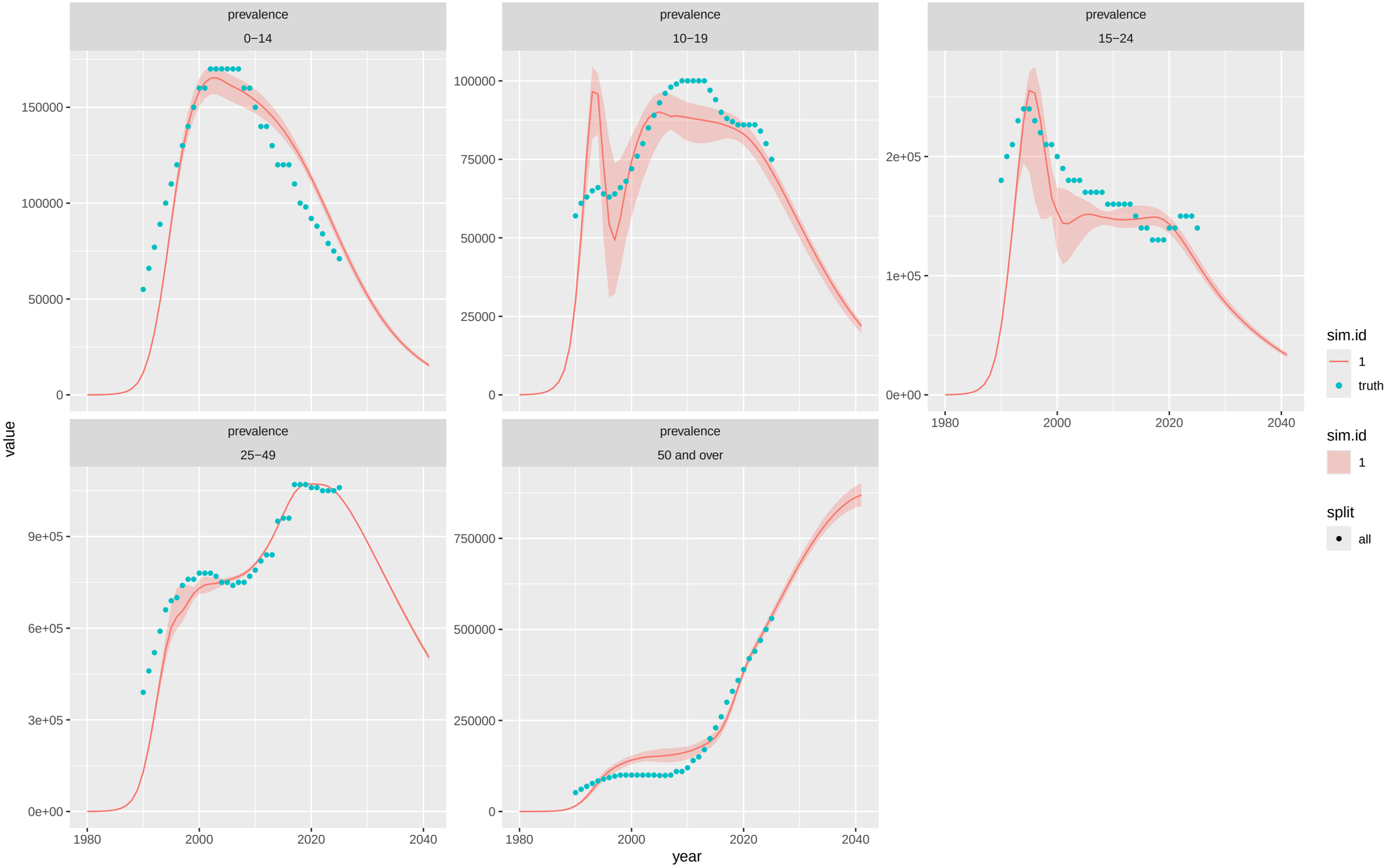
Tanzania: fit (merged Step 2 simset)



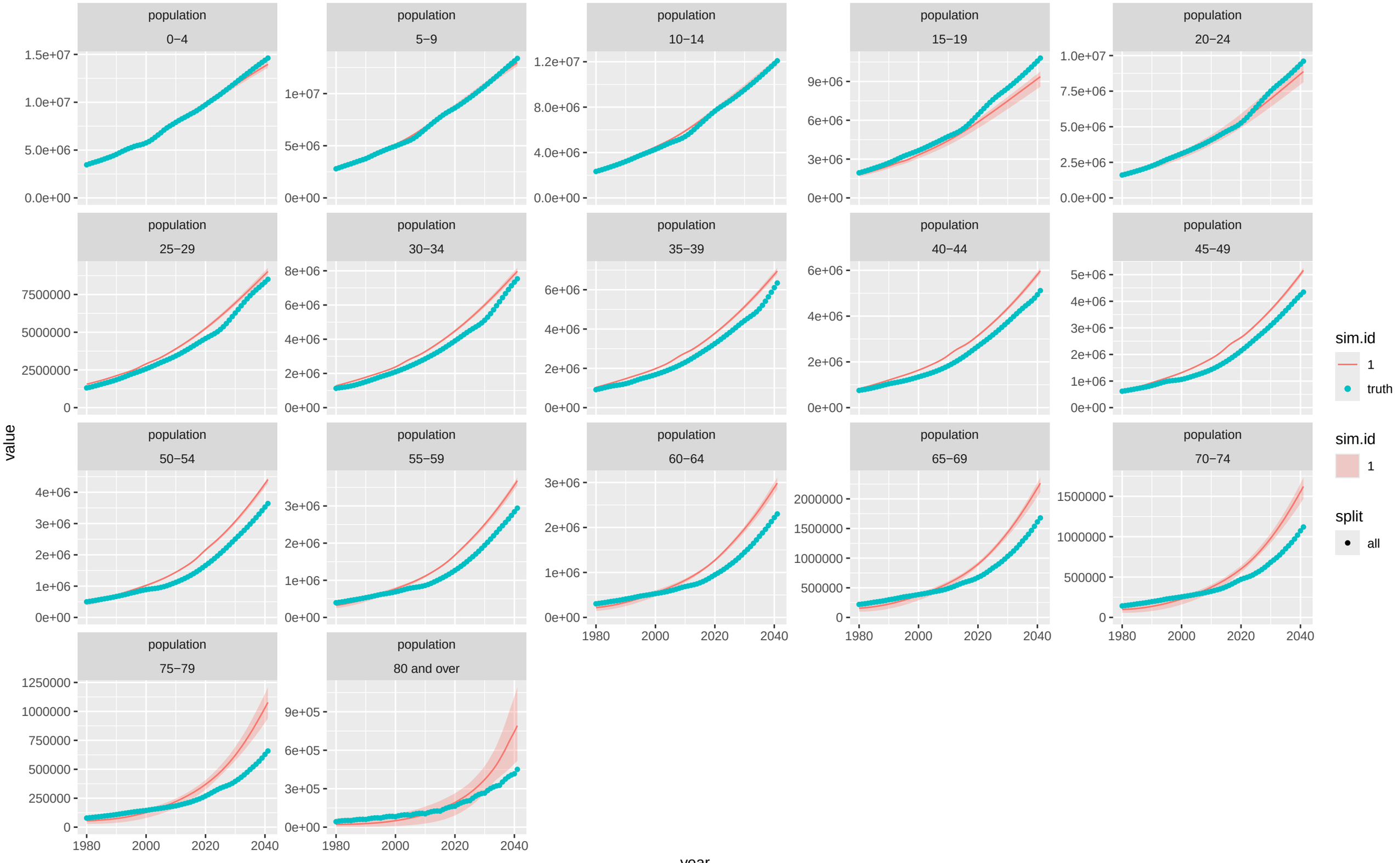
Tanzania: incidence.by.age (merged Step 2 simset)



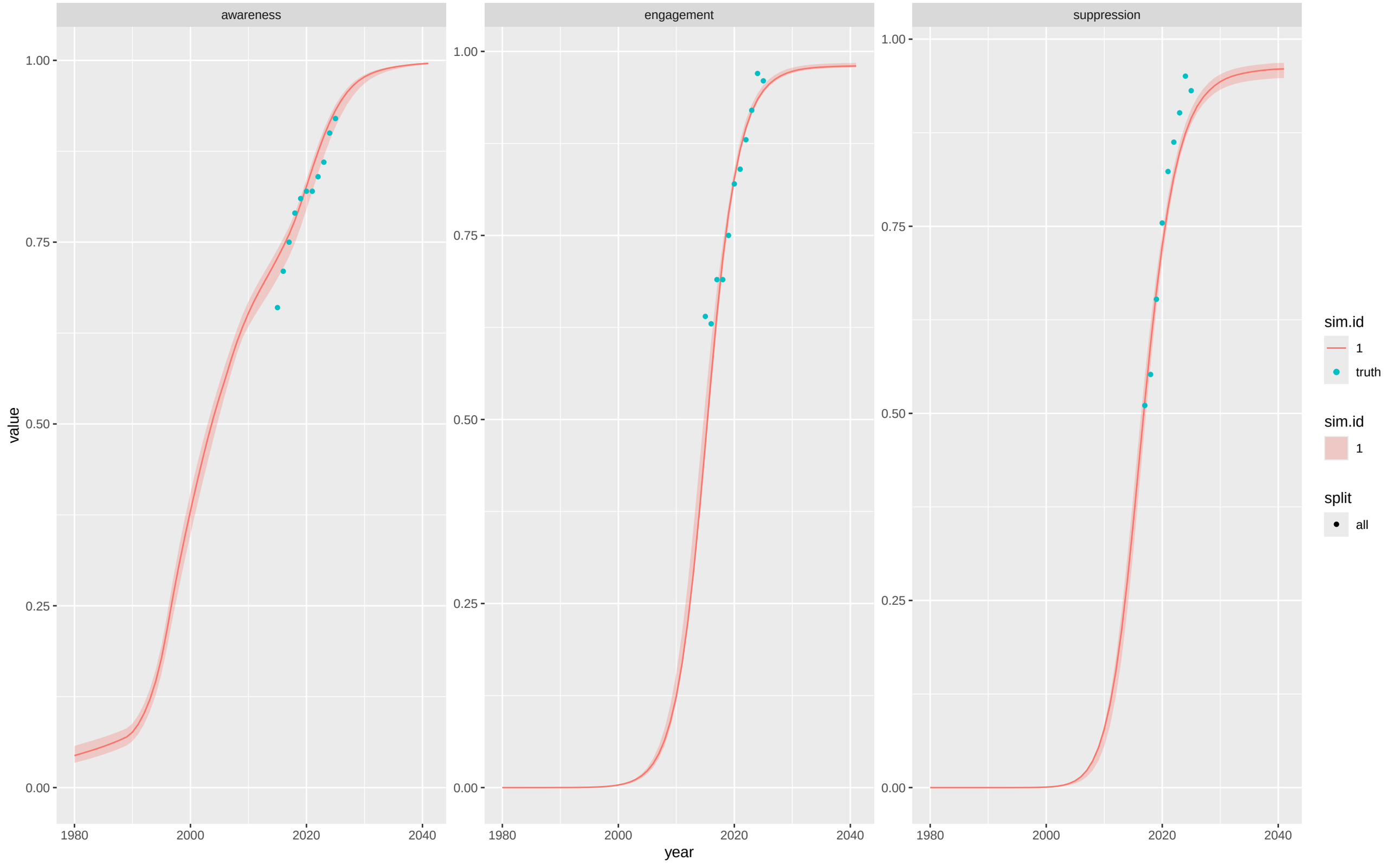
Tanzania: prevalence.by.age (merged Step 2 simset)



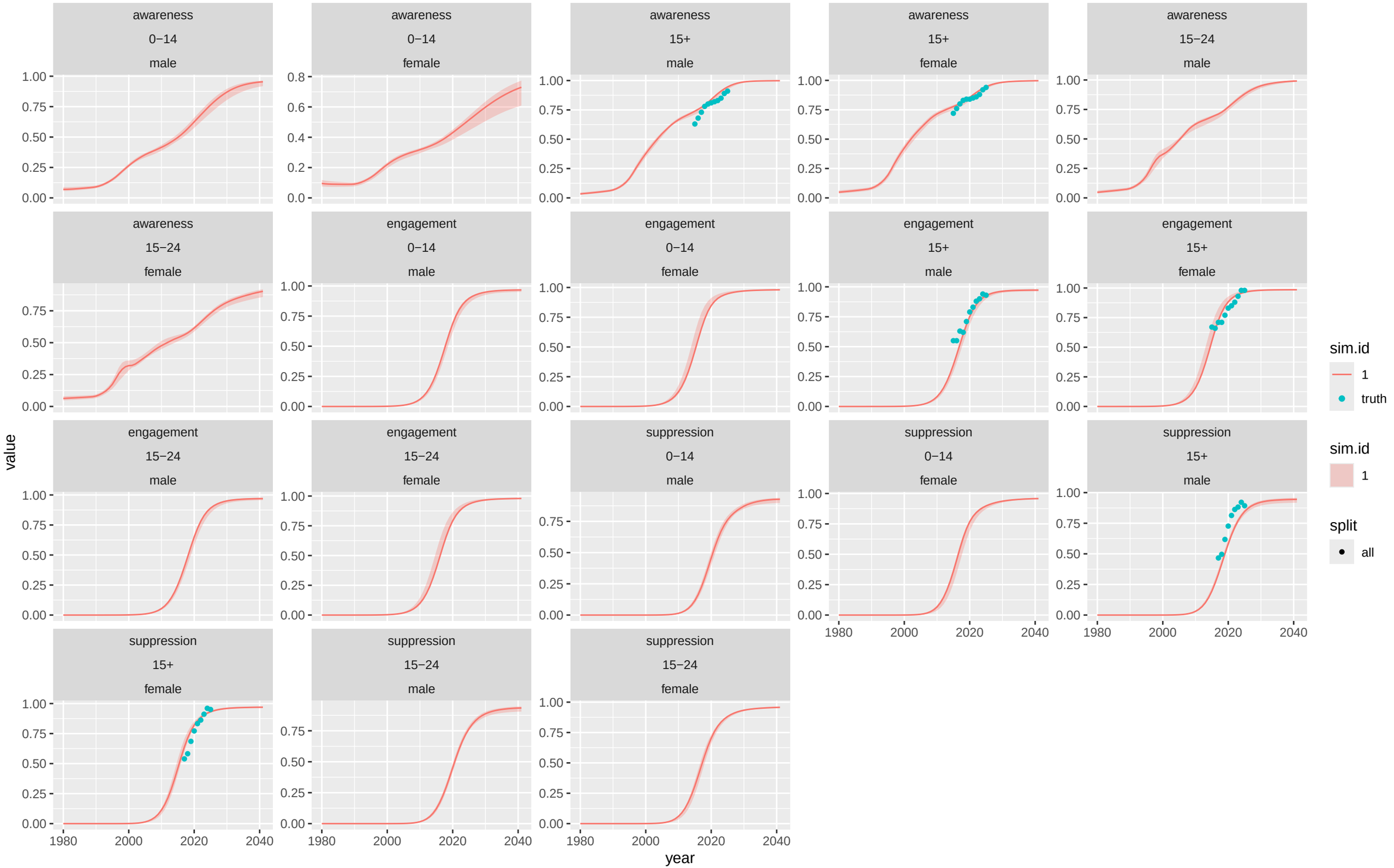
Tanzania: population.by.age (merged Step 2 simset)



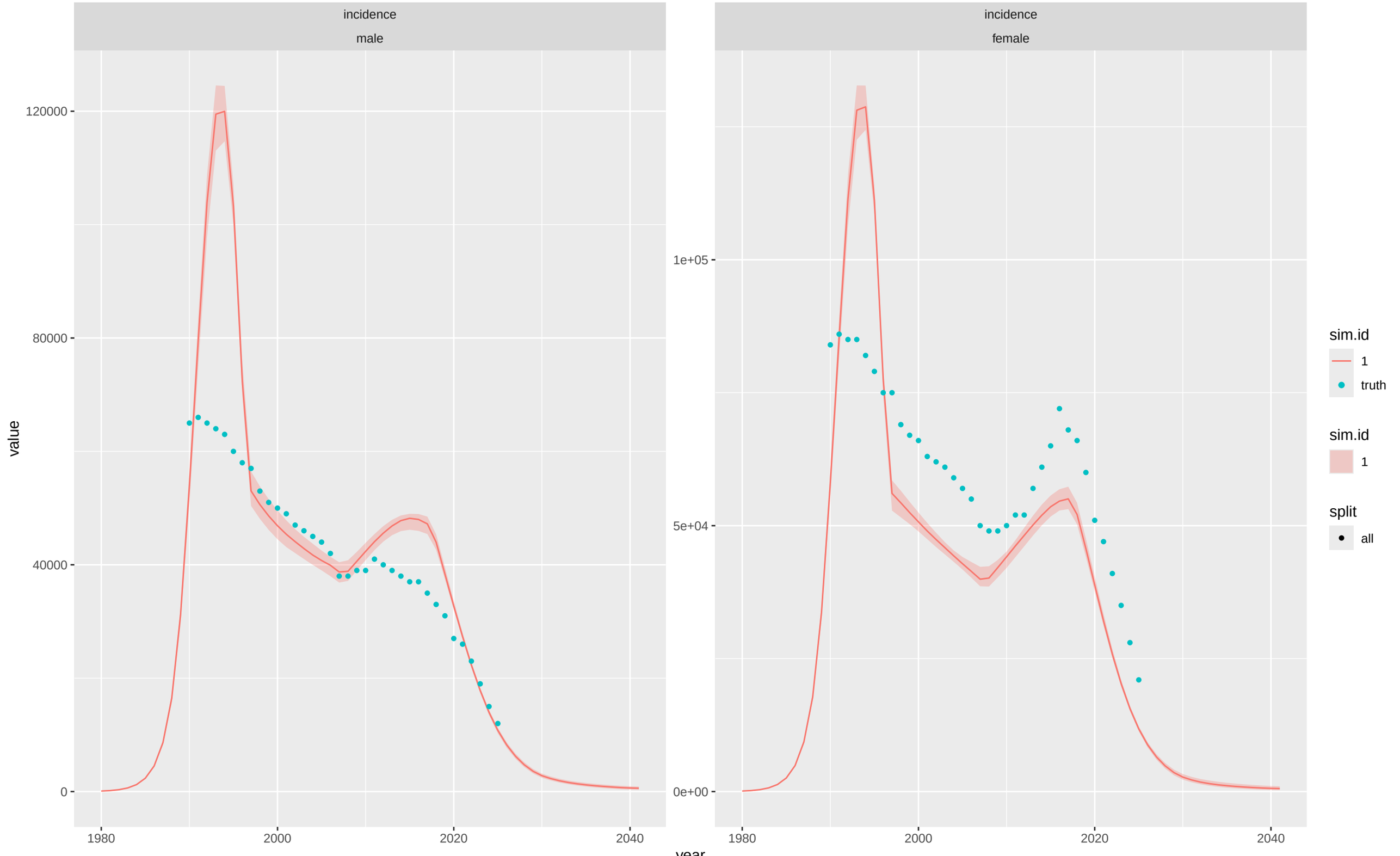
Tanzania: cascade (merged Step 2 simset)



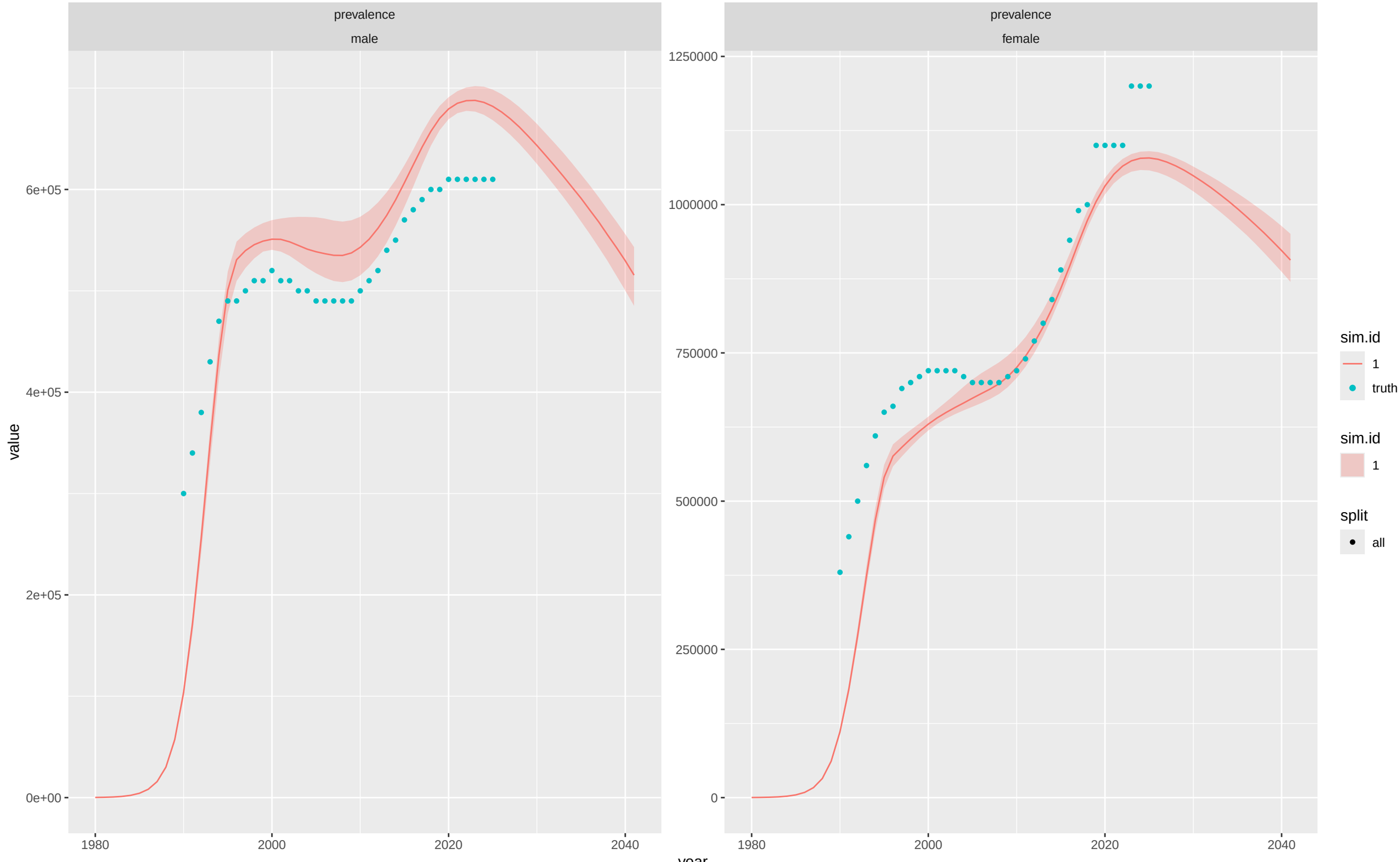
Tanzania: cascade.by.age.sex (merged Step 2 simset)



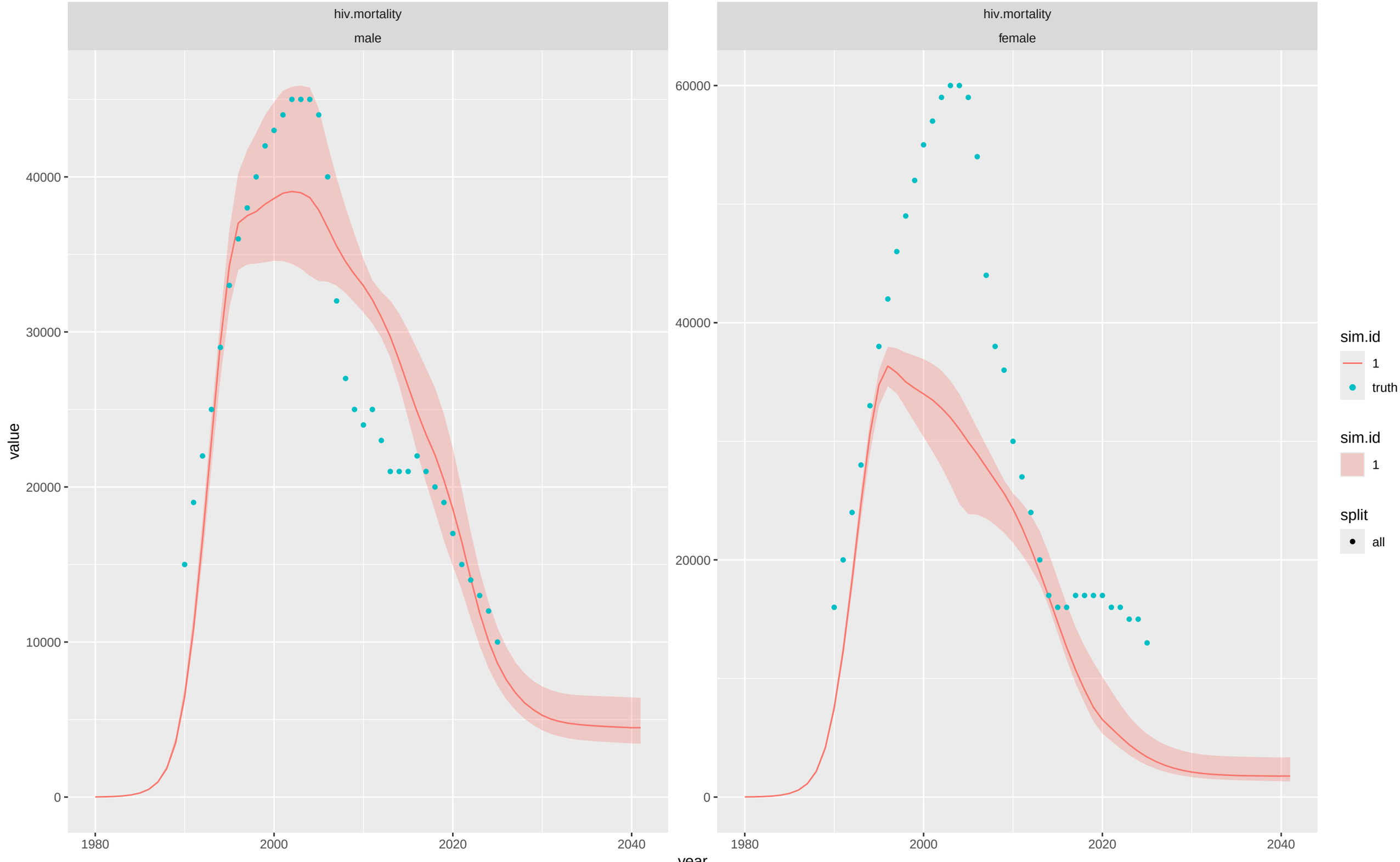
Tanzania: incidence.by.sex (merged Step 2 simset)



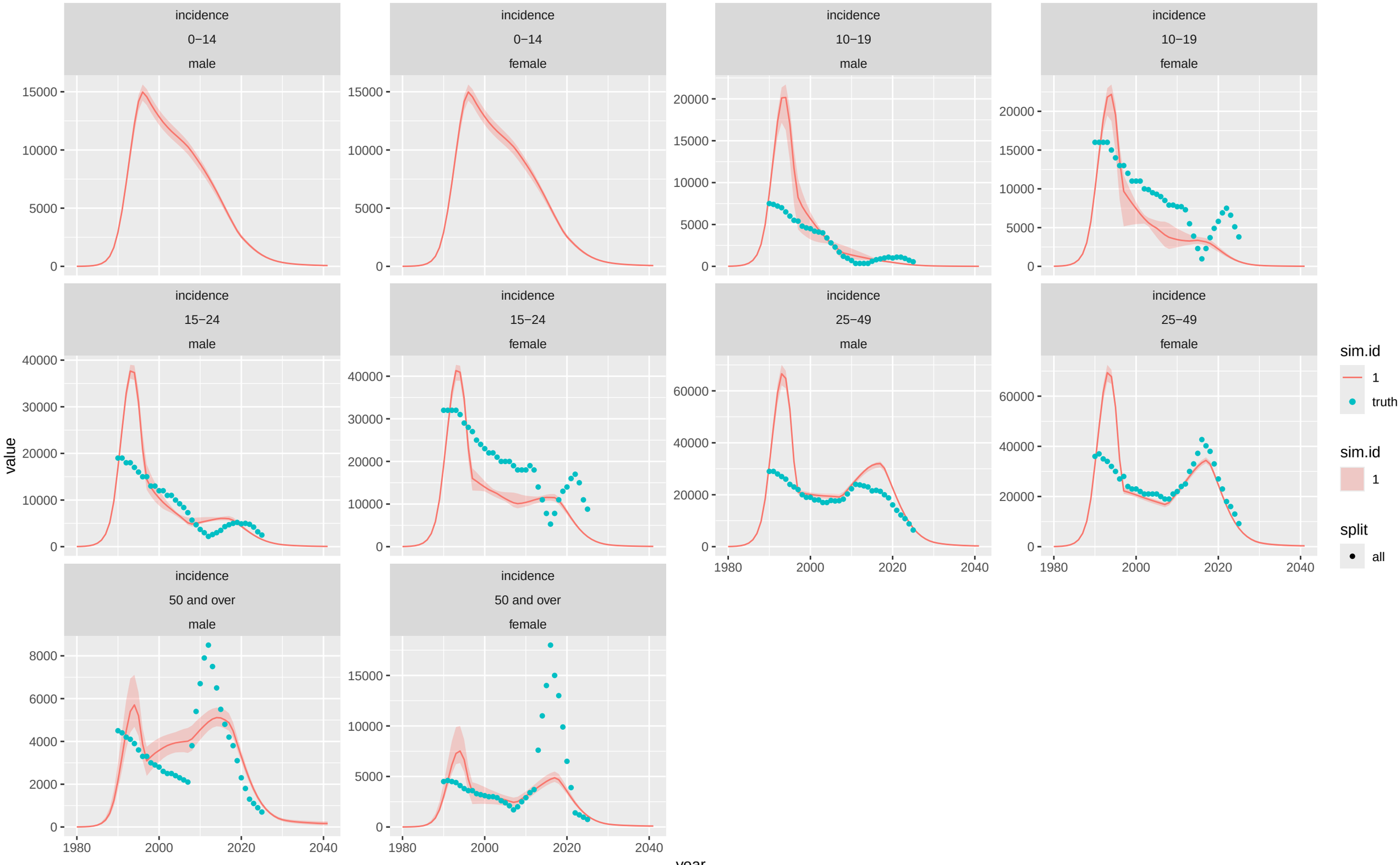
Tanzania: prevalence.by.sex (merged Step 2 simset)



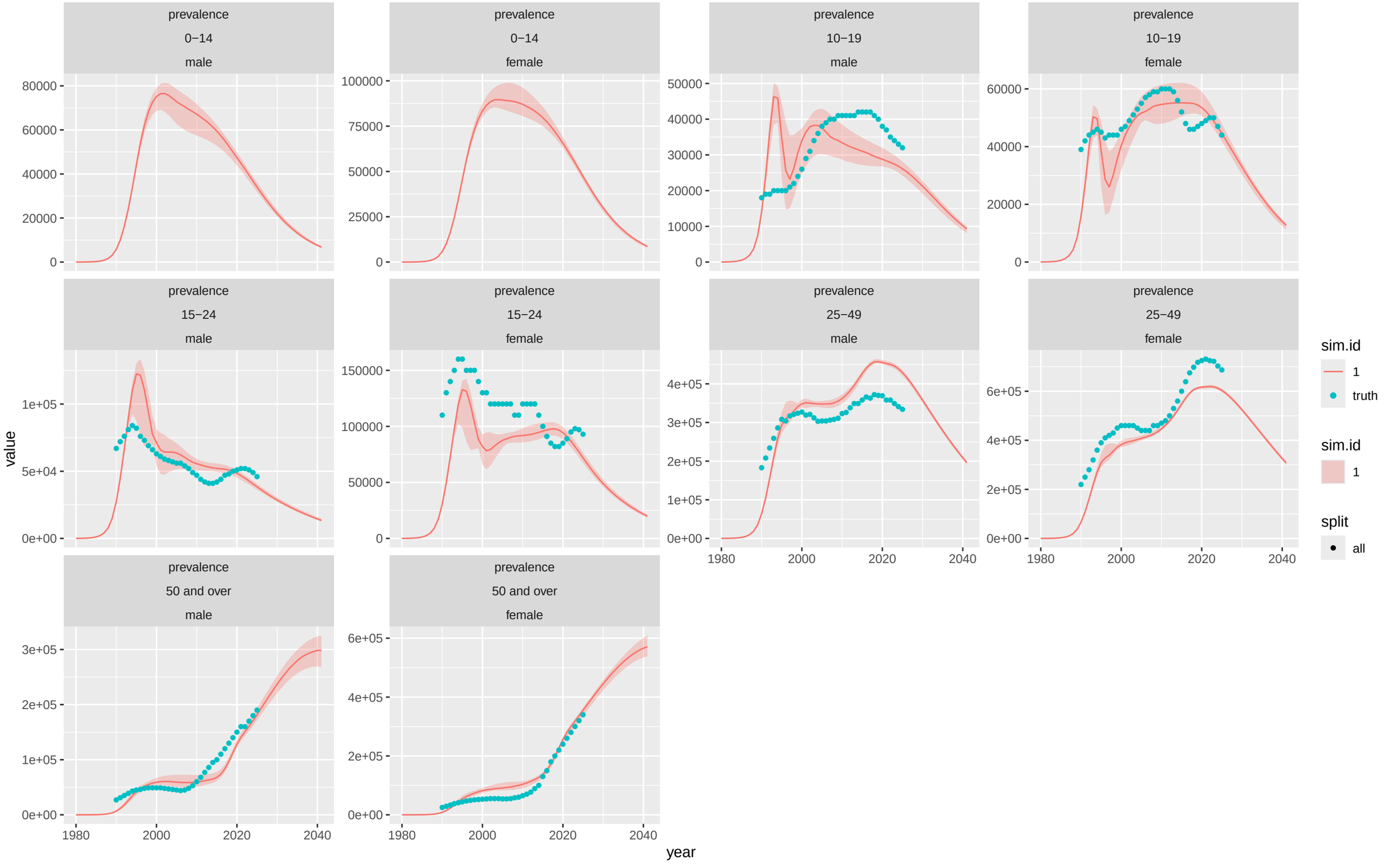
Tanzania: hiv.mortality.by.sex (merged Step 2 simset)



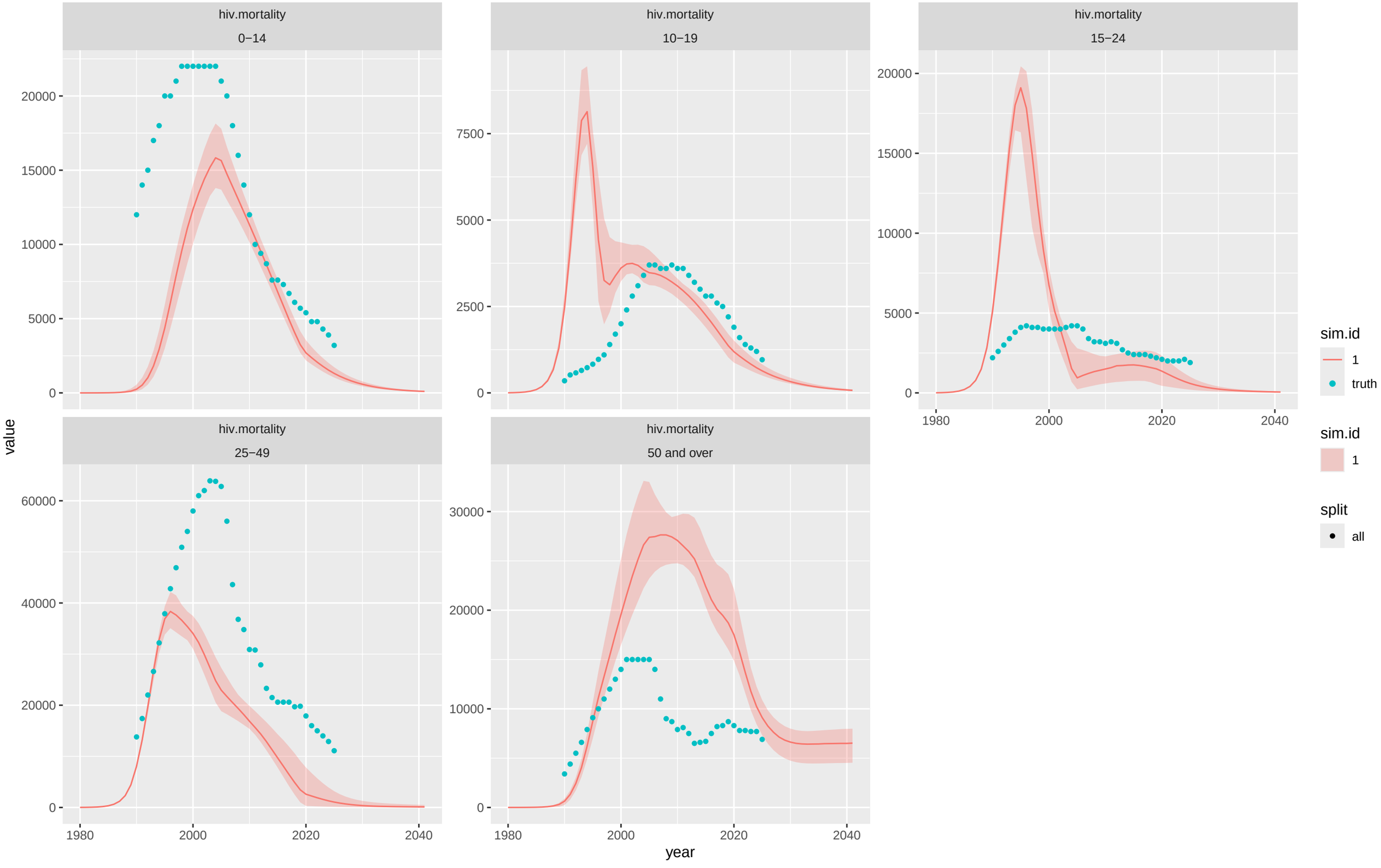
Tanzania: incidence.by.age.sex (merged Step 2 simset)



Tanzania: prevalence.by.age.sex (merged Step 2 simset)



Tanzania: hiv.mortality.by.age (merged Step 2 simset)



Tanzania: hiv.mortality.by.age.sex (merged Step 2 simset)

